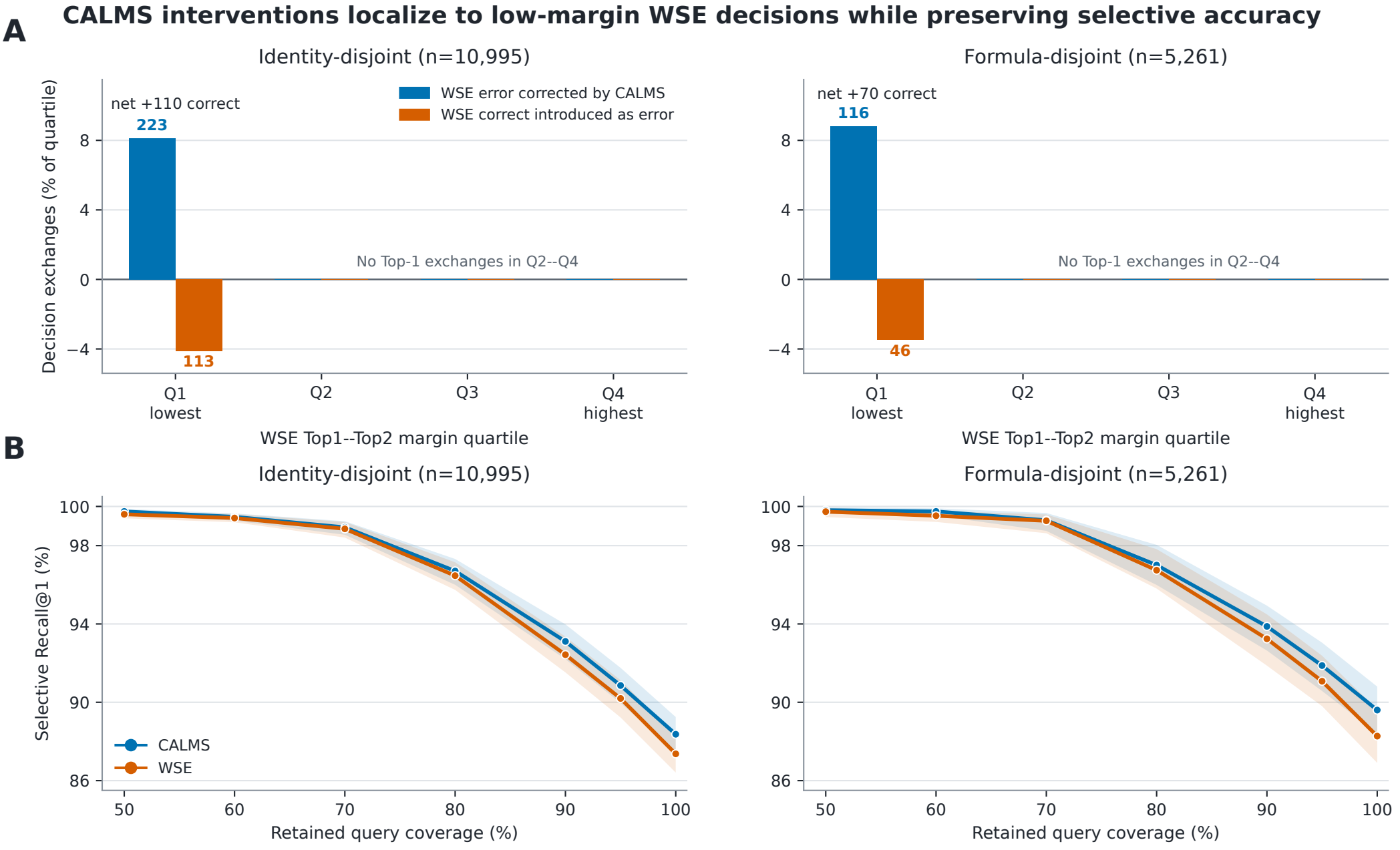

A, Top-1 exchanges from WSE to frozen CALMS v1.0 (internal Full23 v1.3.2), stratified by WSE margin quartile; bar labels are query counts and bar heights are percentages within quartile. B, closed-set selective Recall@1 (shading: 95% formula-cluster bootstrap intervals). GNPS Gold/Silver is consumed development evidence and every query contains its true candidate; these curves do not evaluate no-match/open-set rejection, false-match FDR, or a deployable acceptance threshold.